Yash Patel

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EDUCATION

University of Michigan, Ph.D. in Statistics, GPA: 4.00 **Sep 2021 – Dec 2025**
Selected Courses: Computation and Optimization Methods, Monte Carlo Methods, Systems for Generative AI, LLMs and Transformers, Causal Inference, Uncertainty Quantification in Modeling, Regression Analysis

Princeton University, A.B. in Mathematics **Sep 2014 – Jun 2018**
Certificates: Applications of Computing, Statistics and ML
Selected Courses: Topology, Real Analysis, Complex Analysis, Theoretical ML (Graduate), Fairness in ML (Graduate), Machine Learning/Pattern Recognition (Graduate), Stochastic Systems, Computer Vision

SUMMARY

ML research engineer experienced in **large-scale LLM post-training**, **3D reconstruction & real-time rendering**, and **OpenGL**/**C++**/**Python**. Research expertise in **ML for PDEs**, **uncertainty quantification** (variational inference & conformal prediction), **robust decision-making**, **control theory**, and **convex optimization**.

EXPERIENCE

Harmonic, Research Engineer **Jan 2026 – Present**

- Running **large-scale agentic post-training on 500+ GPUs** with miles (**Megatron-LM** training, **SGLang** rollouts) to train models that **formally verify C and Verilog code in Lean**, including building the **environments and evals**.
- Owned and improved the stability of the **expert-iteration training pipeline** for the **100B+ parameter** policy/value Monte Carlo graph search model in Aristotle.

Meta, Senior Software Engineer (**IC5**) *Managers: Albert Pozo, Brian Cabral* **Jul 2018 – Sep 2021**

- Designed and implemented a **novel real-time (72 FPS) dynamic object reconstruction** algorithm for 300k+ vertex meshes in Unity HLSL/C#.
- Implemented **real-time (72 FPS)** point cloud, dense mesh, and TSDF (KinectFusion) scene **reconstruction & rendering on HMDs and lenticular displays with C++/OpenGL/GLES/OpenCL**.
- Implemented a **gaze-correction model** in PyTorch and optimized it via Qualcomm SNPE and QAT to **run at 30 FPS on a Qualcomm SoC** for Portal. Added PyTorch-JIT-to-Caffe2 translation for quantized nodes.
- Added distributed rendering with Docker, RabbitMQ, and Kubernetes to Manifold camera (`https://github.com/facebook/facebook360_dep`). Reduced depth estimation time by 30% with a novel "Gaussian funnel" technique.

INDUSTRY INTERNSHIPS

Anthropic (Research Engineer, AI Safety Fellowship) *Advisor: Daniel Freeman* **Nov 2025 – Jan 2026**

- Built an embodied robotics evaluation harness in MuJoCo for the Frontier Red Team; selected as **one of 32 researchers out of 2,000+ applicants**.

Waymo (Data Science PhD Intern, Simulations) *Advisor: Aman Sinha* **Jun 2025 – Sep 2025**

- Implemented an ADMM-based **distributed convex optimization** algorithm in C++ for **importance sampling** of **rare events**, achieving a **20x speedup** in the simulation pipeline.

Bose (Machine Learning Research Co-op) *Advisor: Russell Izadi* **Jan 2025 – Jun 2025**

- Implemented **SAC** and **PPO** for adaptive-FIR noise cancellation (PyTorch). Developed a novel transformer-based Wiener filter adaptation that **outperforms FxLMS by 10% in noise reduction (dB)**.

Amazon (Software Engineering Intern) **Jun 2017 – Aug 2017**

- Built a Java Spring MVC debugging service for Kiva Picking Optimization team. Deployed globally via AWS.

ACADEMIC RESEARCH INTERNSHIPS

Columbia University (Research Intern) *Advisor: Abdulrahmen El-Sayed* **May 2015 – Sep 2015**

- Developed agent-based models of how enrollment in exercise-coaching programs affects self-efficacy among sexual minority populations: `https://github.com/yashpatel5400/SexualEqualityABM`

Princeton Plasma Physics Lab (Research Intern) *Advisor: Ilya Dodin* **Jun 2013 – Jan 2014**

- Developed FDTD numerical simulations in C++/Python of the Vlasov equation (see `https://arxiv.org/abs/1006.3717`) to study plasma evolution: `https://yashpatel5400.github.io/files/cos.mp4`

Rutgers University (Research Intern) *Advisor: Michael Shiflett* **Jun 2012 – Aug 2012**

- Investigated the role of axonal guidance in the manifestation of social withdrawal by studying social behavior in mice with NRP2 gene mutations. Prepared brain slices, set up mice trials, and annotated/analyzed data.

PUBLICATIONS

Journal Papers

- "Improved Microplastic Identification from Simultaneously Collected Photothermal Infrared and Raman Spectra Using Multiview Conformal Prediction", Parham RL, Ochoa Rivera E, Ayala AM, Clough ME, **Patel Y**, McNeil AJ, Tewari A, Ault AP. *ACS Measurement Science Au*, 2026.
- "Distribution-Free Robust Predict-Then-Optimize in Function Spaces", **Patel Y**, Tewari A. *SIAM/ASA Journal on Uncertainty Quantification*, 2026 (to appear).

Conference Papers

- "Conformal Contextual Robust Optimization", **Patel Y**, Rayan S, Tewari A. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024. **Oral Presentation**
- "Variational Inference with Coverage Guarantees in Simulation-Based Inference", **Patel Y**, McNamara D, Loper J, Regier J, Tewari A. *International Conference on Machine Learning (ICML)*, 2024
- "Conformal Prediction for Ensembles: Improving Efficiency via Score-Based Aggregation", Ochoa Rivera E*, **Patel Y*** (* equal contribution), Tewari A. *Neural Information Processing Systems (NeurIPS)*, 2025.
- "Continuum Transformers Perform In-Context Learning by Operator Gradient Descent", **Patel Y***, Mishra A*, Tewari A. *International Conference on Learning Representations (ICLR)*, 2026.
- "Conformal Robust Control of Linear Systems", **Patel Y**, Rayan S, Tewari A. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026.

Preprints / In Submission

- "A Greedy PDE Router for Blending Neural Operators and Classical Methods", Rayan S, **Patel Y**, Tewari A.
- "Operator Learning for Schrödinger Equation: Unitarity, Error Bounds, and Time Generalization", **Patel Y***, Subedi U*, Tewari A.
- "Goldilocks: Optimal Importance Sampling with Constraints for Rate Estimation", **Patel Y**, Sinha A.

Workshop Papers

- "Non-Parametric Conformal Distributionally Robust Optimization", **Patel Y**, Cao G, Tewari A. *ICML 2024 Workshop on Structured Probabilistic Inference & Generative Modeling*
- "Diffusion Models for Probabilistic Deconvolution of Galaxy Images", Li Y, Xue Z, **Patel Y**, Regier J. *ICML Machine Learning for Astrophysics Workshop*, 2023
- "RL Boltzmann Generators for Conformer Generation in Data-Sparse Environments", **Patel Y**, Tewari A. *NeurIPS Machine Learning in Structural Biology (MLSB) Workshop*, 2022
- "Scalable Bayesian Inference for Finding Strong Gravitational Lenses", **Patel Y**, Regier J. *NeurIPS Machine Learning and the Physical Sciences (ML4PS) Workshop*, 2022

Patents

- "Holographic Calling for Artificial Reality", AP Pozo, J Virskus, G Venkatesh, K Li, SC Chen, A Kumar, R Ranjan, BK Cabral, SA Johnson, W Ye, MA Snower, **Y Patel**. *US Patent App. 17/360,693*

AWARDS

- NSF GRFP Honorable Mention (2020, 2022)
- Outstanding First-Year Ph.D. Student Award (2022)
- Outstanding GSI Team Award (2022)
- Siemens Westinghouse National Semifinalist (2014)

INVITED PRESENTATIONS

- *Conformal Contextual Robust Optimization*. "Contextual Stochastic Optimization: From Data to Decisions" Workshop at 2026 Banff International Research Station (BIRS)
- *Learning Evolution Operator of the Time-Dependent Schrödinger Equation*. SciFM 2025 Conference
- *Continuum Transformers Perform In-Context Learning by Operator Gradient Descent*. "In-context Learning for PDEs and Inverse Problems" Workshop at 2026 SIAM Conference on Uncertainty Quantification.

SERVICE

Reviewer: NeurIPS, ICLR, AISTATS, Reliable ML from Unreliable Data Workshop @ NeurIPS, IEEE Conference on Decision and Control (CDC), IEEE Control Systems Letters

Teaching: STATS 315 (Statistics & AI): Created HWs and lab materials for first offerings of this course and taught labs, STATS 485 (Capstone Seminar), STATS 504 (Practice and Communication in Applied Statistics), STATS 250 (Introduction to Statistics and Data Analysis)

SKILLS

Python, C++, C, C#, CUDA, PyTorch, OpenCL, OpenGL, Unity, Docker, Kubernetes, Git, Java, Bash, LaTeX